

Natural-Language versus Symbolic Messaging for Ground-Aerial Open-Vocabulary Search in a Toy World: What the Message Says Matters, the Room-Level Reference Costs

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<https://github.com/mihir-chauhan/nl-ground-aerial-ovs-search>

Abstract—Heterogeneous robot teams are increasingly coordinated through natural language, but it is rarely tested whether the language format itself helps compared with a fixed message schema. We build a small scripted grid-world testbed for open-vocabulary object search by a ground robot and an aerial robot in which only the message channel changes: templated English sentences read by a rule-based parser (object, colour, nearby landmark, room name) versus fixed-schema tuples (class id and coordinates), at the same message budget and on identical layouts. Over four query tiers and 5 seeds, the natural-language team was not faster than the symbolic team: pooled over tiers it needed 1.28 [0.42, 2.16] more steps (paired mean, 95% bootstrap interval), with a small gain in success of 2.0 [0.4, 3.8] percentage points. The difference depends on the query as hypothesised: language was slower for exact and synonym queries and reached higher success for relational queries. A decomposition explains both effects. Attribute and relation content helps regardless of format, since a schema with two extra fields behaves identically to a sentence with coordinates, and locating objects by room name instead of coordinates costs 2.28 [1.79, 2.80] steps. Under message corruption both channels degrade and the symbolic one does not degrade less. Communication of either kind matters clearly more than the choice of format: the step difference to the same team without a channel is -8.46 (symbolic) and -7.18 (language). The study is a CPU-only toy with a templated language channel; it tests the information-format hypothesis, not the behaviour of language models.

Index Terms—multi-robot systems, heterogeneous teams, open-vocabulary object search, inter-robot communication, natural language, frontier exploration

I. Introduction

A ground robot and an aerial robot make a natural search team. The aerial robot covers area quickly but sees coarsely and cannot look under cover; the ground robot is slow but can identify an object at close range. When the target is specified in free text (“the red mug”, “the mug near the sofa”), the two must also exchange what

they see. Recent systems let robots do this in natural language, often through large language models [1]–[4]. Language is attractive because it is open-ended: a sentence can mention a colour, a relation or a room without anyone having designed a field for it.

Whether that open-endedness pays off in task performance is a separate question from whether a full language stack works. A sentence and a tuple can carry the same facts, and a sentence typically refers to places loosely (“in the kitchen”) where a tuple gives coordinates. The research question of this study is therefore narrow: in a simulated grid world, does natural-language communication between a ground robot and an aerial robot improve open-vocabulary object search success and time-to-find compared with structured symbolic messaging and with non-communicating frontier exploration?

The approved brief fixed three hypotheses before any experiment. H1: the natural-language team reaches higher success and fewer steps than the symbolic team at equal message budget. H2: the advantage grows as queries become more open-vocabulary and vanishes or reverses for exact class names. H3: the advantage shrinks under message corruption.

We test them in a deliberately small setting: a procedurally generated grid world, a simulated open-vocabulary detector, scripted frontier agents, and a templated language channel with a rule-based parser standing in for a language model. This keeps every factor other than the channel identical and lets us pair episodes across systems. It also bounds what can be claimed: the study is about the information format of messages, not about language models.

Findings. H1 is not supported for time-to-find: pooled over query tiers the language team is slightly slower than the symbolic team, with a small success gain. H2 is supported in direction: the language-minus-symbolic difference moves in favour of language from closed to open query tiers (interaction -2.51 [-4.21 , -0.86] steps). H3 as stated presupposes an advantage that we did not observe;

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under corruption both channels lose and the gap does not move toward the symbolic channel. A decomposition shows that what helps is the attribute and relation content, which a slightly richer schema carries equally well, while the room-level reference of our sentences is what costs time.

Contributions. (i) A reproducible testbed in which only the inter-robot message channel changes. (ii) A controlled comparison of templated language and fixed-schema messaging across four query tiers with paired layouts. (iii) A content-versus-format decomposition and ablations on message budget, corruption, aerial resolution, detector noise and team composition, including a coordination failure of our own design that the ablation exposed.

II. Related Work

Open-vocabulary object search. Zero-shot semantic navigation scores where to explore with vision-language models: VLFM builds a language-grounded value map over frontiers [5], OneMap keeps a reusable open-vocabulary feature map for multi-object search [6], and HOV-SG organises open-vocabulary features into a floor, room and object scene graph for language-grounded navigation [7]. HM3D-OVON broadens object-goal navigation to goals given in free-form language [8]. The perception side relies on open-vocabulary detectors such as OWL-ViT [9]. Our detector is a simulated stand-in for this text-similarity scoring, and our SPL-style metric follows the standard evaluation recommendations for embodied navigation [10].

Multi-robot exploration and search. Frontier-based exploration and its multi-robot variants remain the reference family, benchmarked for example in Explore-Bench [11]. For semantic search, GoalSwarm coordinates several aerial robots on a shared top-down semantic map [12], and COMRES-VLM lets a vision-language model coordinate frontier assignment over shared maps [13]. These systems share maps or state directly; the message format is not the object of study.

Language between heterogeneous robots. Cladera et al. present air-ground collaboration for language-specified missions, where a ground and an aerial robot share information under intermittent communication [1]. MHRC uses language models for decentralised collaboration among heterogeneous robots [2]; Chen et al. compare centralised and decentralised language-model dialogue for multi-robot planning with attention to token budgets [3]; CoELA builds embodied agents that plan and communicate in natural language when communication is costly [4]; and Choe et al. let heterogeneous robots request and offer help in natural language [14]. These works evaluate complete language stacks. They do not hold the agents fixed and swap only the message format.

What to communicate. In multi-agent reinforcement learning, a survey organises communication by what is sent, to whom and under which constraints [15], and Hill et

al. compare an engineered intention message with an end-to-end learned protocol [16]. Our study is in this spirit but with hand-written channels: we compare two engineered formats and separate message content from the way a location is referred to.

Gap. To our knowledge the closest systems [1], [2] report end-to-end results for language-coordinated heterogeneous teams. We did not find a controlled comparison of natural-language against fixed-schema messages for ground-aerial open-vocabulary search at matched budget on paired layouts; this toy study addresses the information-format part of that question only.

III. Method

The testbed is built so that the message channel is the only thing that differs between the compared teams. Agents, sensing, planning and random numbers are shared. Everything is scripted; there is no learned component and no language model.

A. World and queries

Each episode draws a 30×30 grid that is split recursively into 4 to 8 rectangular rooms separated by walls with one or two doors. Rooms carry names from a fixed list (kitchen, lounge, office, ...). A room is covered by a roof with probability 0.3; the aerial robot cannot see into covered rooms. Each room holds one or two landmarks (sofa, table, bed, shelf, desk), and the world holds 12 small objects: the target, two distractors and nine fillers. Objects belong to 20 classes organised in 5 groups of visually similar classes (for example mug, bottle, glass, bowl), have one of 6 colours, and are described as near a landmark when one lies within Chebyshev distance two. The floor plan (walls, room names, roofs) is known to both robots; object positions are not.

A query is drawn from one of four tiers of increasing distance from a closed class vocabulary: exact (“the mug”), synonym (“the cup”, a word outside the class list), attribute (“the red mug”, with two mugs of other colours present), and relational (“the mug near the sofa”, with two mugs away from any sofa present). In the exact and synonym tiers the two distractors are other classes of the target’s similarity group, so that the amount of clutter is the same in all tiers.

B. Simulated open-vocabulary detector

Real open-vocabulary detectors such as OWL-ViT [9] score image regions against free text. We replace the image side by a symbolic percept and keep the text-similarity side. Every class, synonym, colour and landmark word has a fixed random 64-dimensional unit embedding. Classes of one group share a common component, and a synonym is a perturbed copy of its class vector, so that a synonym query matches its class less sharply than the exact name does. A percept is a triple (label, colour, landmark). Its score against a query is the mean, over the slots the query

mentions, of the cosine between the query word and the perceived word; a slot the percept does not know scores one half, and an absent landmark scores zero. Gaussian noise is added to the score (standard deviation 0.05 for the ground robot and 0.10 for the aerial robot).

C. Robots

The ground robot moves one cell per step on the 4-connected free grid, senses the cells of its own room within Chebyshev radius 3, and perceives label, colour and position exactly. Objects whose noisy score reaches 0.75 become its own candidates; it walks to the nearest one and verifies it at close range, which takes 3 steps and is exact.

The aerial robot moves 2 cells per step over walls and senses a square footprint of radius 4 except under roofs. Its perception is coarse: the label is replaced by another class of the same group with probability 0.2, the colour is perceived with probability 0.7, and positions are quantised to blocks of 3×3 cells, which also makes its landmark relations approximate. It cannot verify from altitude. Once its own coverage is complete it falls back to descending on its best candidates, which costs 10 steps each.

Both robots explore with nearest-frontier coverage in the spirit of frontier exploration [11]: the goal is the nearest cell not yet sensed, kept until it is sensed, with a cost penalty on cells of the room the teammate last announced. When no unsensed cell remains the coverage map is reset and a new pass starts with fresh detector noise. An episode succeeds when either robot verifies the target within 100 steps.

D. Message channels

A robot may send one message every $K = 5$ steps. A message carries at most $M = 2$ sightings and, when it changed, the sender’s intent (where it is heading). The aerial robot reports every object whose class score reaches 0.6, best first; it does not filter on colour or relation, because its perception of those is unreliable, and leaves that judgement to the receiver. The ground robot acts on its own sightings and sends only intents. The receiver turns each reported sighting into a candidate region, re-scores what the message says against the query with the detector’s scorer, drops entries below 0.7, and ranks the rest by score minus travel distance divided by 50. A region is resolved once the ground robot has sensed all of it. The channels differ only in how a sighting record is written and read:

NL messaging (ours). The speaker fills an English template, for example “i am heading to the office. i see a red mug near the sofa in the kitchen and a mug in the lab”. Colour and landmark are mentioned when perceived. The location is the room name. The listener is a rule-based keyword parser (a mock language model): it scans the words, opens an entry at each class word, attaches colour, landmark and room words, and discards entries without

a class or a location. The candidate region is the whole named room.

Symbolic messaging (baseline). The message is a list of fixed-schema tuples: (G, x, y) for intent and (S, class_id, x, y) for a sighting. The schema has no slot for colour or relation, so the receiver scores those slots as unknown. The candidate region is the block around the coordinates.

Decomposition variants. To separate what a message says from how it says it we add three channels: NL + coordinates (the same sentence with “at x y” inserted), Symbolic + attributes (two more fields: colour id and landmark id), and Symbolic + confidence (one more field: the sender’s own full score, which the receiver uses instead of re-scoring).

Information match and mismatch. All channels encode the same sighting record. NL and the basic schema are deliberately not matched in content: NL carries colour and relation but only a room-level location, while the schema carries a block-level location but no colour or relation. This is the trade-off the hypotheses are about. NL + coordinates and Symbolic + attributes carry identical information; because the template and parser are lossless without corruption, these two are behaviourally identical in clean conditions, and we report both only to make that point explicit. Budgets are matched in messages and sightings per message, not in tokens: an NL message uses about twice as many tokens (words) as a symbolic one uses fields (Table I).

Corruption. For H3 each NL room word is replaced by another room name with probability p and then each word is dropped with probability p . The symbolic analogue replaces each coordinate pair by a random cell with probability p and then erases each field with probability p ; a tuple with an erased required field is discarded.

IV. Experimental Setup

Scale. This is a small CPU study in a toy world with scripted agents. All experiments together took about half an hour on two threads of a shared machine.

Systems. NL messaging is the method under test. The baselines, all reimplemented here on the same agents, are: Symbolic messaging; Frontier team, no comm (the same two frontier robots without a channel, in the tradition of multi-robot frontier exploration [11]); Single ground frontier; Random-walk team (both robots move randomly, the ground robot still walks to and verifies its own detections); and Oracle sharing, in which sighting records are passed losslessly at every step with no entry limit. The oracle bounds the channel, not the task: it still shares only what the aerial robot perceived.

Tasks and seeds. The main comparison runs 40 episodes per query tier and seed, with 5 seeds, for every system. Ablation sweeps use a mixed task whose episodes cycle through the four tiers, with 80 episodes per seed. Layout, sensor noise and corruption noise depend only on seed, tier and episode index, so all systems face the same worlds

and the same percepts (common random numbers), and episodes can be paired.

Metrics. Success within the step cap; steps to find, set to the cap when the target is not found; an SPL-style score [10] (success times the shortest ground path from start to target divided by the steps taken); tokens per episode (words for NL, fields for symbolic); and overlap, the Jaccard index of the cells ever sensed by the two robots.

Statistics. Tables give mean and standard deviation over seeds. Because seeds are few, the decisive comparisons are paired at the episode level: for two systems we take the per-episode difference on identical layouts and report its mean with a 95% percentile bootstrap interval (10000 resamples of episodes) and a sign-flip permutation test.

Development and tuning. Separate development seeds were used for debugging and to choose the step cap, which was set so that the single ground robot succeeds in roughly half of the episodes; with a cap twice as long every frontier system succeeds almost always and success carries no information. Detector, sensing and channel parameters were fixed on plausibility before the evaluation seeds were run and were not changed afterwards. No parameter is tuned per system.

V. Results

A. Sanity ordering

Table I and Fig. 1 show the expected ordering of the reference systems in every tier: the random-walk team almost never finds the target, the single ground robot succeeds in about half of the episodes, the two-robot team without communication is clearly better (the aerial robot helps only through its slow fallback verification), and the communicating teams are better again. Against the team without a channel, symbolic messaging changes the steps to find by -8.46 $[-10.20, -6.74]$ and NL messaging by -7.18 $[-8.78, -5.62]$ steps pooled over tiers (Table II). The oracle is at most a few steps ahead of the symbolic channel (-2.07 $[-3.17, -1.01]$ pooled), so the headroom above both real channels is small in this world.

B. H1: language versus symbolic at equal message budget

Not supported for time-to-find; small gain in success. Pooled over the four tiers, the NL team takes 1.28 $[0.42, 2.16]$ more steps than the symbolic team, an interval that excludes zero on the side of the symbolic channel. Its success rate is higher by 2.0 $[0.4, 3.8]$ percentage points (Table III), an interval that barely excludes zero. A seed-level paired t -test on the pooled task agrees on the direction for steps (NL 64.12 against symbolic 62.84 steps, $p = 0.015$, 5 seeds). Both effects are small against the seed-to-seed spread in Table I, and the NL channel uses about twice as many tokens. We therefore do not find that the language channel yields both higher success and lower time-to-find; on time it loses slightly.

C. H2: dependence on the query tier

Supported in direction, with a reversal for closed queries. For exact class names the NL team is slower by 2.98 $[2.03, 4.01]$ steps and for synonyms by 2.08 $[0.95, 3.19]$ steps; both intervals exclude zero. For attribute queries the step difference is 1.27 $[-0.94, 3.48]$ and for relational queries -1.23 $[-3.38, 0.78]$; both intervals include zero. The contrast between the open tiers (attribute, relational) and the closed tiers (exact, synonym) is -2.51 $[-4.21, -0.86]$ steps in favour of language on the open tiers. In success (percentage points), the NL team gains 7.0 $[3.0, 11.0]$ on relational queries and 4.0 $[-0.5, 8.5]$ on attribute queries (the latter interval includes zero), and nothing on the closed tiers.

Two details qualify the hypothesis. First, the synonym tier behaves like the exact tier: a synonym makes the detector less sharp for both channels alike, and the language channel has no extra way to exploit it, because both receivers match the reported class against the query with the same scorer. Second, the language channel never becomes clearly faster than the symbolic one; on the open tiers it roughly breaks even on time and wins on success.

A further observation concerns the symbolic channel alone. On relational queries its success is lower than that of the team without communication (-5.5 $[-11.0, -0.5]$ percentage points): class-only reports send the ground robot to same-class distractors that it would otherwise not have visited within the step cap. Content-free sightings can hurt.

VI. Ablations and Analysis

A. Content versus format

The decomposition channels (Table II, Fig. 2) separate the two differences between our sentences and the basic schema.

Content. Adding coordinates to the sentence gives a channel that is faster than the basic schema by -1.00 $[-1.71, -0.34]$ steps pooled, with the gain concentrated on relational (-2.48 $[-4.53, -0.56]$) and attribute (-1.54 $[-3.42, 0.33]$) queries and exactly zero on exact and synonym queries, where there is no colour or relation to exploit. Adding colour and landmark fields to the schema gives the same numbers in every cell: with a lossless template and parser the two channels deliver identical records, so the receiver behaves identically. Sending only the sender’s confidence as one extra field recovers a similar pooled gain (-0.98 $[-1.72, -0.25]$). The benefit therefore belongs to the information, not to the language format.

Location reference. Replacing coordinates by a room name in an otherwise identical sentence costs 2.28 $[1.79, 2.80]$ steps pooled, and the cost is present in every tier. The ground robot has to search the named room instead of walking to a block of cells.

Together these account for the main result: our NL channel adds content that helps on open queries and pays a room-level localisation cost on all queries.

TABLE I

Main comparison: mean $\pm$ standard deviation over 5 seeds. The four query tiers have 40 episodes per seed; “all” pools those episodes; “mixed” is a separate set of 80 episodes per seed that cycles through the tiers. All baselines are reimplemented on the same agents. Tokens are words (NL) or tuple fields (symbolic); the oracle is not a message channel and counts none.

Method	Task	success $\uparrow$	steps $\downarrow$	spl $\uparrow$	tokens $\downarrow$	overlap $\downarrow$	n
Oracle sharing	exact	0.85 ± 0.08	58.21 ± 2.70	0.48 ± 0.05	0.00 ± 0.00	0.31 ± 0.02	5
Single ground frontier	exact	0.53 ± 0.10	80.17 ± 5.61	0.24 ± 0.06	0.00 ± 0.00	0.00 ± 0.00	5
Random-walk team	exact	0.03 ± 0.01	98.83 ± 0.62	0.01 ± 0.00	0.00 ± 0.00	0.30 ± 0.04	5
Symbolic messaging	exact	0.86 ± 0.04	58.56 ± 1.56	0.48 ± 0.04	33.98 ± 1.28	0.31 ± 0.01	5
Frontier team, no comm	exact	0.79 ± 0.04	72.24 ± 4.26	0.35 ± 0.04	0.00 ± 0.00	0.39 ± 0.03	5
NL messaging	exact	0.84 ± 0.04	61.54 ± 1.40	0.44 ± 0.03	69.14 ± 2.31	0.33 ± 0.01	5
Oracle sharing	synonym	0.78 ± 0.12	62.74 ± 6.19	0.46 ± 0.08	0.00 ± 0.00	0.32 ± 0.03	5
Single ground frontier	synonym	0.52 ± 0.06	77.59 ± 6.27	0.26 ± 0.05	0.00 ± 0.00	0.00 ± 0.00	5
Random-walk team	synonym	0.02 ± 0.02	99.47 ± 0.76	0.00 ± 0.01	0.00 ± 0.00	0.31 ± 0.03	5
Symbolic messaging	synonym	0.79 ± 0.12	64.21 ± 4.88	0.44 ± 0.06	34.98 ± 0.71	0.31 ± 0.03	5
Frontier team, no comm	synonym	0.77 ± 0.09	70.84 ± 6.95	0.38 ± 0.05	0.00 ± 0.00	0.38 ± 0.02	5
NL messaging	synonym	0.78 ± 0.11	66.29 ± 4.82	0.42 ± 0.06	70.89 ± 1.28	0.33 ± 0.02	5
Oracle sharing	attribute	0.82 ± 0.07	61.98 ± 3.07	0.46 ± 0.03	0.00 ± 0.00	0.31 ± 0.03	5
Single ground frontier	attribute	0.43 ± 0.09	81.96 ± 3.45	0.20 ± 0.04	0.00 ± 0.00	0.00 ± 0.00	5
Random-walk team	attribute	0.03 ± 0.02	98.59 ± 1.10	0.01 ± 0.01	0.00 ± 0.00	0.34 ± 0.03	5
Symbolic messaging	attribute	0.77 ± 0.07	65.37 ± 4.21	0.41 ± 0.04	37.96 ± 2.40	0.35 ± 0.03	5
Frontier team, no comm	attribute	0.73 ± 0.09	73.64 ± 3.69	0.33 ± 0.05	0.00 ± 0.00	0.40 ± 0.04	5
NL messaging	attribute	0.80 ± 0.08	66.65 ± 3.93	0.40 ± 0.05	73.94 ± 3.53	0.33 ± 0.03	5
Oracle sharing	relational	0.82 ± 0.06	60.12 ± 2.86	0.47 ± 0.03	0.00 ± 0.00	0.29 ± 0.04	5
Single ground frontier	relational	0.49 ± 0.06	76.25 ± 2.52	0.25 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	5
Random-walk team	relational	0.03 ± 0.03	99.54 ± 0.54	0.00 ± 0.01	0.00 ± 0.00	0.33 ± 0.04	5
Symbolic messaging	relational	0.76 ± 0.05	63.21 ± 3.44	0.44 ± 0.04	38.24 ± 0.90	0.33 ± 0.03	5
Frontier team, no comm	relational	0.81 ± 0.07	68.47 ± 3.73	0.39 ± 0.05	0.00 ± 0.00	0.39 ± 0.04	5
NL messaging	relational	0.82 ± 0.06	61.98 ± 3.76	0.45 ± 0.04	73.96 ± 0.76	0.30 ± 0.03	5
Oracle sharing	all	0.82 ± 0.05	60.77 ± 1.75	0.47 ± 0.03	0.00 ± 0.00	0.31 ± 0.01	5
Single ground frontier	all	0.49 ± 0.03	78.99 ± 1.71	0.24 ± 0.02	0.00 ± 0.00	0.00 ± 0.00	5
Random-walk team	all	0.02 ± 0.01	99.11 ± 0.26	0.01 ± 0.00	0.00 ± 0.00	0.32 ± 0.01	5
Symbolic messaging	all	0.79 ± 0.04	62.84 ± 1.26	0.44 ± 0.03	36.29 ± 0.80	0.32 ± 0.01	5
Frontier team, no comm	all	0.77 ± 0.04	71.30 ± 1.99	0.36 ± 0.02	0.00 ± 0.00	0.39 ± 0.01	5
NL messaging	all	0.81 ± 0.04	64.12 ± 0.59	0.43 ± 0.03	71.98 ± 0.82	0.32 ± 0.01	5
Oracle sharing	mixed	0.85 ± 0.04	58.57 ± 3.22	0.49 ± 0.04	0.00 ± 0.00	0.29 ± 0.02	5
Single ground frontier	mixed	0.53 ± 0.05	77.60 ± 3.10	0.26 ± 0.03	0.00 ± 0.00	0.00 ± 0.00	5
Random-walk team	mixed	0.02 ± 0.02	99.09 ± 0.68	0.01 ± 0.01	0.00 ± 0.00	0.33 ± 0.02	5
Symbolic messaging	mixed	0.85 ± 0.03	59.29 ± 2.67	0.49 ± 0.04	34.97 ± 0.93	0.30 ± 0.02	5
Frontier team, no comm	mixed	0.78 ± 0.05	71.05 ± 2.32	0.37 ± 0.03	0.00 ± 0.00	0.40 ± 0.02	5
NL messaging	mixed	0.85 ± 0.04	60.81 ± 2.62	0.47 ± 0.04	69.94 ± 2.41	0.30 ± 0.02	5

TABLE II

Paired episode-level difference in steps to find, system A minus system B on identical layouts (negative: A is faster), with 95% bootstrap interval in brackets. Each tier pools 5 seeds of 40 episodes; “all” pools the four tiers.

Comparison (A – B)	exact	synonym	attribute	relational	all
NL – Symbolic	2.98 [2.03, 4.01]	2.08 [0.95, 3.19]	1.27 [–0.94, 3.48]	–1.23 [–3.38, 0.78]	1.28 [0.42, 2.16]
NL + coordinates – Symbolic	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	–1.54 [–3.42, 0.33]	–2.48 [–4.53, –0.56]	–1.00 [–1.71, –0.34]
Symbolic + attributes – Symbolic	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	–1.54 [–3.42, 0.33]	–2.48 [–4.53, –0.56]	–1.00 [–1.71, –0.34]
Symbolic + confidence – Symbolic	0.00 [0.00, 0.00]	0.18 [–0.05, 0.58]	–0.94 [–3.07, 1.21]	–3.16 [–5.26, –1.19]	–0.98 [–1.72, –0.25]
NL – NL + coordinates	2.98 [2.03, 4.01]	2.08 [0.95, 3.19]	2.81 [1.61, 4.08]	1.25 [0.70, 1.85]	2.28 [1.79, 2.80]
Oracle – Symbolic	–0.34 [–2.27, 1.50]	–1.47 [–3.17, 0.10]	–3.38 [–5.64, –1.20]	–3.09 [–5.86, –0.46]	–2.07 [–3.17, –1.01]
Symbolic – No comm	–13.68 [–17.00, –10.42]	–6.63 [–10.23, –3.01]	–8.28 [–11.69, –4.98]	–5.26 [–9.01, –1.56]	–8.46 [–10.20, –6.74]
NL – No comm	–10.70 [–13.96, –7.50]	–4.55 [–8.00, –1.10]	–7.00 [–9.82, –4.29]	–6.49 [–9.70, –3.31]	–7.18 [–8.78, –5.62]

B. H3: message corruption

Not supported as stated. H3 assumed a language advantage that would shrink under corruption. On the mixed task there is no advantage to begin with: without corruption the NL-minus-symbolic step difference is 1.51 [0.26, 2.75]. Corruption hurts both channels (Fig. 3): at the highest rate the NL team loses 9.68 [7.43, 11.93] steps and the symbolic team 11.22 [8.57, 13.87] steps relative to their clean versions, so fidelity matters for both. But the gap does not move toward the symbolic channel. At rates 0.1, 0.2 and 0.4 the difference is -0.68 [–2.90, 1.58], -0.86 [–3.36, 1.62] and -0.03 [–2.46, 2.37], and the shift relative to the clean gap is -2.19 [–4.33, –0.04], -2.38 [–4.92,

$0.24]$ and -1.54 [–4.21, 1.05]: toward language if anything, with intervals that mostly include zero. A plausible reason is redundancy in our corruption models: a dropped colour or landmark word leaves a usable sentence, and a wrong room is discarded after a bounded search, whereas an erased tuple field discards the whole sighting and a wrong coordinate pair sends the robot to an arbitrary cell. This depends on how corruption is defined for each format and should not be read as a general robustness claim.

C. Message budget, aerial resolution, detector noise

The sign of the NL-minus-symbolic difference is stable across the sensitivity sweeps (Fig. 4). With a message ev-

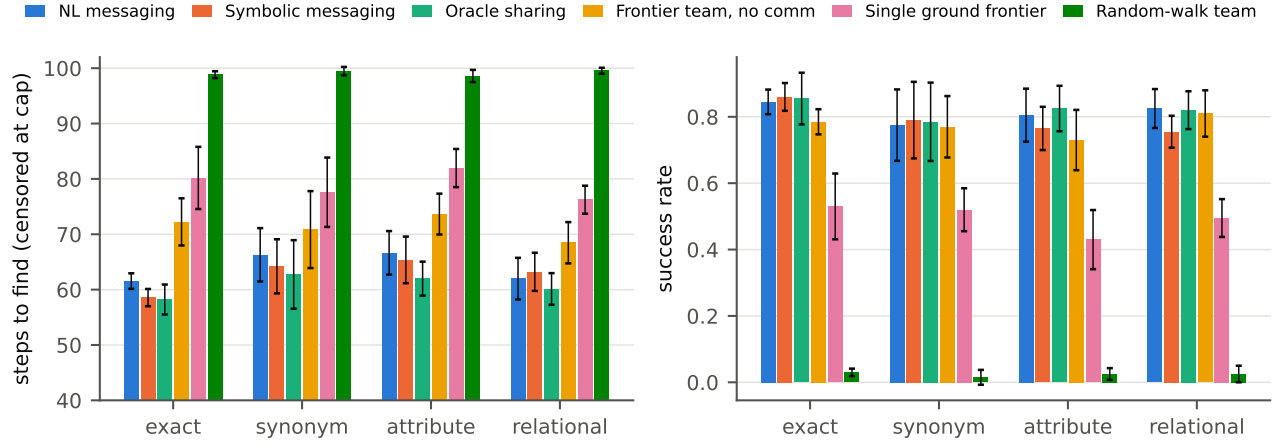


Fig. 1. Steps to find (left, lower is better) and success rate (right) per query tier. Bars are means over 5 seeds, whiskers one standard deviation.

TABLE III

Paired episode-level difference in success rate in percentage points, A minus B, with 95% bootstrap interval (positive: A succeeds more often).

Comparison (A - B)	exact	synonym	attribute	relational	all
NL - Symbolic	-1.5 [-4.0, 0.5]	-1.5 [-4.0, 0.5]	4.0 [-0.5, 8.5]	7.0 [3.0, 11.0]	2.0 [0.4, 3.8]
NL + coordinates - Symbolic	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	5.5 [1.5, 10.0]	7.5 [4.0, 11.5]	3.2 [1.9, 4.8]
Symbolic + attributes - Symbolic	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	5.5 [1.5, 10.0]	7.5 [4.0, 11.5]	3.2 [1.9, 4.8]
Symbolic + confidence - Symbolic	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.5 [-2.5, 5.5]	8.0 [4.0, 12.0]	2.4 [1.0, 3.9]
Symbolic - No comm	7.5 [2.0, 13.0]	2.0 [-4.0, 8.0]	3.5 [-2.5, 9.5]	-5.5 [-11.0, -0.5]	1.9 [-0.9, 4.8]
NL - No comm	6.0 [1.0, 11.0]	0.5 [-5.0, 6.0]	7.5 [2.0, 13.0]	1.5 [-2.5, 6.0]	3.9 [1.4, 6.5]

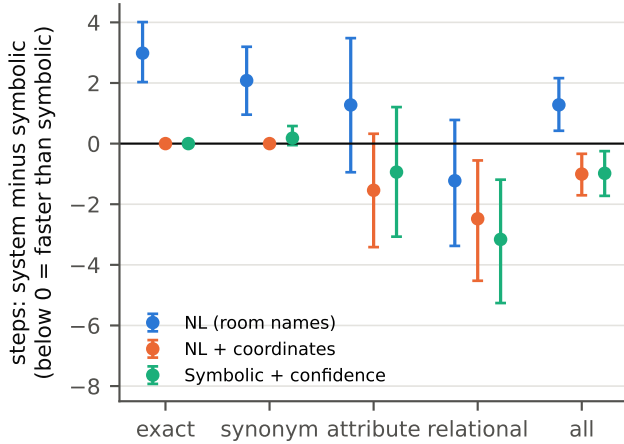


Fig. 2. Content and format decomposition: paired step difference to the basic symbolic channel, per tier, with 95% bootstrap intervals. The Symbolic + attributes channel coincides with NL + coordinates and is not drawn.

ery step it is 1.27 [-0.03, 2.52] steps, with a message every 20 steps 1.00 [-0.02, 1.97]. With exact aerial positions (block of one cell) it is 1.75 [0.52, 2.97] and with blocks of five cells 1.89 [0.49, 3.33]; coarser aerial positions do not erase the coordinate advantage at this scale. Without

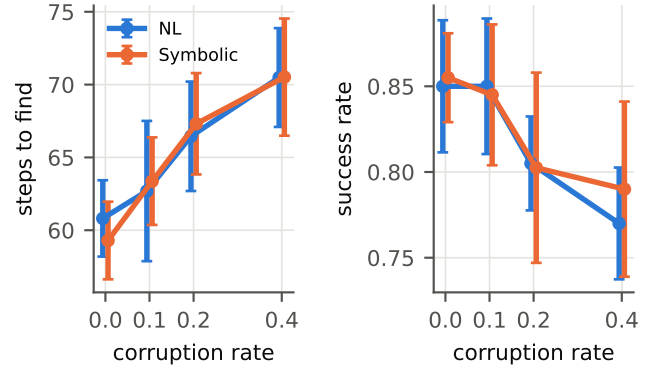


Fig. 3. Message corruption on the mixed task (mean and standard deviation over 5 seeds of 80 episodes).

detector noise and label confusion it is 0.97 [-0.29, 2.20], with doubled noise 1.02 [-0.28, 2.30]. Several of these intervals touch zero; none favours language.

D. Team composition and a coordination failure

Table IV varies the team. With the mutual room-avoidance rule of the main configuration, two ground robots or two aerial robots that communicate do far worse than the same teams without a channel. Tracing episodes showed a symmetric livelock: two identical robots that

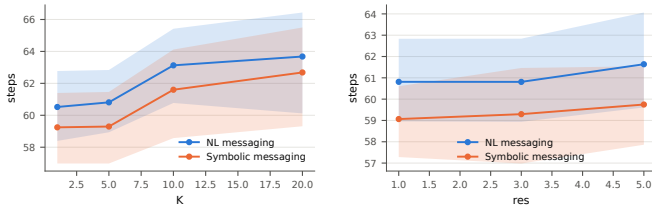


Fig. 4. Sensitivity on the mixed task: steps to find against the message interval K (left) and the aerial quantisation block (right).

TABLE IV

Team composition and coordination rule on the mixed task (5 seeds of 80 episodes). “Priority rule”: only the second robot avoids the room its teammate announced.

Method	success $\uparrow$	steps $\downarrow$	tokens $\downarrow$	overlap $\downarrow$	n
No comm, two aerial robots	0.56 ± 0.06	87.67 ± 2.32	0.00 ± 0.00	1.00 ± 0.00	5
NL, priority rule	0.81 ± 0.03	63.16 ± 2.21	67.03 ± 1.25	0.36 ± 0.02	5
Symbolic, priority rule	0.82 ± 0.04	60.91 ± 2.21	33.56 ± 0.41	0.35 ± 0.02	5
Symbolic messaging	0.85 ± 0.03	59.29 ± 2.67	34.97 ± 0.93	0.30 ± 0.02	5
NL, two aerial robots, priority rule	0.58 ± 0.06	86.78 ± 2.28	117.30 ± 2.05	0.99 ± 0.00	5
NL, two ground robots, priority rule	0.77 ± 0.06	63.86 ± 4.11	47.62 ± 2.59	0.35 ± 0.03	5
NL, two ground robots	0.28 ± 0.04	88.37 ± 1.23	189.54 ± 3.59	1.00 ± 0.00	5
NL, two aerial robots	0.12 ± 0.05	98.42 ± 0.74	200.27 ± 1.95	1.00 ± 0.00	5
No comm, two ground robots	0.53 ± 0.05	77.60 ± 3.10	0.00 ± 0.00	1.00 ± 0.00	5
NL messaging	0.85 ± 0.04	60.81 ± 2.62	69.94 ± 2.41	0.30 ± 0.02	5

start on the same cell announce the same room, both yield, announce the next room, and keep moving in lockstep, which is visible as an overlap of one. The ground-aerial team does not suffer from this because its robots differ in speed and sensing. We added, after seeing this, a priority rule in which only the second robot yields. It removes the livelock: two ground robots exchanging only intents then reach 63.86 steps and 77.3% success, close to the ground-aerial NL team (60.81 steps, 85.0%). For the ground-aerial team the priority rule is slightly worse than the mutual rule for both channels and leaves their order unchanged (Table IV). Two aerial robots gain nothing from intents, since each must sweep the whole open area anyway and verification is their bottleneck.

Two honest readings follow. The heterogeneous team is not much better than a well-coordinated pair of ground robots in this world, so the value of heterogeneity here is modest. And the coordination content of a message (who goes where) mattered at least as much as its format.

VII. Limitations

Templated language, not a language model. The NL channel is a template with a keyword parser. It tests the information-format hypothesis (what a sentence can carry that a fixed schema cannot, and what a room-level reference costs). It says nothing about free-form dialogue, hallucination, ambiguity resolution, or the reasoning a real language model would add or get wrong.

Design choices decide the trade-off. That NL locates by room and the schema by coordinates is our reading of the two formats, taken from the brief. The decomposition shows that the conclusions follow from exactly this choice: a sentence with coordinates and a schema with attribute

fields behave identically. A different template would give a different NL result.

Toy world and simulated perception. The detector is a slot-wise similarity over random word vectors with Gaussian noise, not a vision model; no pretrained text encoder was used. The floor plan is known, localisation is perfect, data association is perfect, objects are static and verification at close range is exact. Effect sizes are specific to this world: one grid size, one object count, one step cap.

Small effects, few seeds. The channel differences are a few steps on episodes of about sixty steps. With 5 seeds, seed-level tests have little power; our conclusions rest on paired episode-level intervals, which treat episodes as independent and ignore between-seed variance. Several per-tier intervals include zero. We ran many comparisons without correction for multiplicity.

Baselines are reimplemented and simple. Frontier exploration is a nearest-unsensed-cell rule on a known map, not a full cost-utility planner on an unknown map. The coordination rule (avoid the room the teammate announced) is naive; it livelocks for co-located identical robots, which we found during the team ablation and report rather than hide. The two-aerial team is limited by the fact that aerial robots verify only as a slow fallback.

Budget matching. Channels are matched in messages and sightings per message, not in tokens or bits; NL uses about twice as many tokens.

What a full study would add. A real language model as speaker and listener, a real open-vocabulary detector on rendered or real imagery, a photorealistic simulator with unknown maps, larger teams, token-matched budgets, more seeds, and corrected tests.

VIII. Conclusion

In a toy ground-aerial open-vocabulary search task where only the message channel changes, templated natural-language messages did not beat fixed-schema symbolic messages on time-to-find; pooled over query types they were slightly slower, with a small gain in success. The gain appears where the hypothesis placed it, on attribute and relational queries, and disappears or reverses for exact and synonym queries. The decomposition locates the causes: attribute and relation content helps in either format, and a room-name reference costs a couple of steps compared with coordinates. Under message corruption both channels degrade and symbolic messaging is not the more robust one under our corruption models. Any communication is worth far more than the choice of format.

For system builders the practical reading is that a language channel earns its place through what it can say that the schema designer did not anticipate, not through being language; where a field can be added, a field does the same job at half the tokens, and precise spatial references should be kept alongside the sentence. Whether real

language models change this picture, through inference or through error, is outside what this study can say and is the natural next experiment.

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